SOP-IT-009 · Cold storage: data and wallets

SOP IT-009 | Status draft | Changed 2026-11-05 | For Anyone who keeps something they cannot replace

Department: IT · **ID:** 009 · **Status:** draft (approved when tagged) · **Owner:** the foundation's stewards · **Last changed:** 2026-11-05

> ***Important. Not legal, tax or investment advice.*** *This page explains how storage fails. It doesn't say what to buy or hold, and it describes nobody's holdings.*

> ***Note. TL;DR.*** *Hot means connected and ready. Cold means it takes a person and a physical step to reach. Cold protects you from what happens over a network, and exposes you to what happens in a room. So think of* ***fire, flood, theft and "I lost it" as four separate failures****, and keep copies in* ***two places, not one safe****. A safe rated for paper isn't rated for a drive.*
>
> ***For wallets, one fact people miss:*** *the backup cards of a seedless card wallet such as Tangem are* ***copies*** *of one key, and* ***each can sign alone****. A stolen card is guarded only by its access code. That is useful, and it is* ***not multisig****.*
>
> ***Test your recovery.*** *A copy you have never restored from is a hope. And* ***never type a seed into a website.***

If you want to know	Go to
Hot and cold for files	Data
Hot and cold for wallets	Wallets
What a Tangem card set really is	Seedless card wallets
What fire does, and what theft does	Four failures
Where to keep things	Two places, not one safe
How to know it works	Test the recovery

Purpose

Give a person the words and the failure list they need before deciding where to keep something they cannot replace, whether that is a file or a key.

Scope

Personal and organisational data, and self-held crypto wallets, explained in general terms. **Not covered:** which product to buy, what any treasury does, tax, inheritance law. How the lab backs up its machines is proposal 019, *Backed up three ways*. Apple account recovery is SOP-COM-002. Encrypted images are SOP-IT-008.

Roles

Role	Does
The holder	decides where things go, keeps their own private list, tests recovery
Anyone helping in person	may read this page with the holder; never sees a seed or an access code

Helper (an agent)	explains and points to sources; **never** reads, holds, types or stores a seed, a code or a key, and never asks for one

Data: hot and cold

How cold	Example	What it gives	What it risks
Hot	iCloud, GitHub, a synced folder	always there, out of the house	an account lock-out; a deletion or a scramble travels with the sync
Warm	a Time Machine drive plugged in at the desk	every file by the hour, with no internet needed	it sits in the same room, and a bad script can reach a mounted drive
Cold	a drive unplugged and kept elsewhere, updated by rotation	out of reach of malware and of the room	only as fresh as the last rotation; forgotten drives fail unnoticed
Coldest	paper	needs no power, no password, no software	a single fire or flood; anyone who finds it can read it

The 3-2-1 rule. Three copies, on two kinds of media, with one away from the house. That is CISA's rule, in its paper *Data Backup Options*, as quoted in proposal 019 on Nov 4, 2026. CISA's PDF refused our fetch on Nov 5, so that quote wasn't re-read. Proposal 019 walks through the test row by row: what passes today, and what a drive and an off-site copy would each add.

Hot and cold help each other. A Time Machine drive is at risk from malware while it is mounted, and a cloud copy is at risk from a lock-out. Proposal 019's table shows no single column stays strong all the way down.

Paper is for the few things that open everything else: the Apple recovery key and your password manager's emergency kit (SOP-COM-002), a FileVault recovery key (SOP-IT-007), an image's password (SOP-IT-008). The test for them is the same: walk through the day every device is gone, and look for a loop.

Wallets: hot and cold

Depth	Hot	Cold
In plain words	The key lives on a phone or a computer that is online. Quick, and open to whatever gets onto that device.	The key lives somewhere offline. Slow on purpose, and the risks become physical.
Technically	The private key is held in software on a networked device, and signing happens there.	The private key stays in an offline device or chip. The transaction goes in, the signature comes out, and the key doesn't leave.

A wallet doesn't hold the coins; the coins are on a public ledger. It holds **the key that can move them**. So everything below is really one question: **who, or what, can sign?**

Four ways to hold a key

	In plain words	Technically	What a thief needs	What losing it means

Seed phrase	A list of ordinary words that can rebuild your whole wallet.	A BIP-39 mnemonic of 12 to 24 words from which keys are derived, with an optional extra passphrase (BIP-39).	the words, and nothing else (plus the passphrase, if one was set)	words gone and devices gone: the wallet is gone
Hardware wallet	A small device that keeps the key inside and signs for you.	The key stays in the device, and it usually shows the transaction on its own screen. Most have a seed as the backup, so the seed is the real target.	the device and its PIN, **or** the seed alone	device gone: restore from the seed. Seed gone too: gone
Multisig	A wallet that takes several keys to move anything, like a safe with two locks and two people.	M of N: a set number of signers out of the total must approve (Ledger Academy). No one person or key can move funds alone (Trezor).	enough keys to reach M	depends on the numbers: 2 of 3 survives losing one key, 3 of 3 doesn't. Trezor adds that losing the extended public keys can stop you sending even with enough private keys
Seedless card wallet	Two or three cards that each hold the same key, with no words to write down.	See below.	**one** card and its access code	one card gone: use another. All gone with no seed: gone

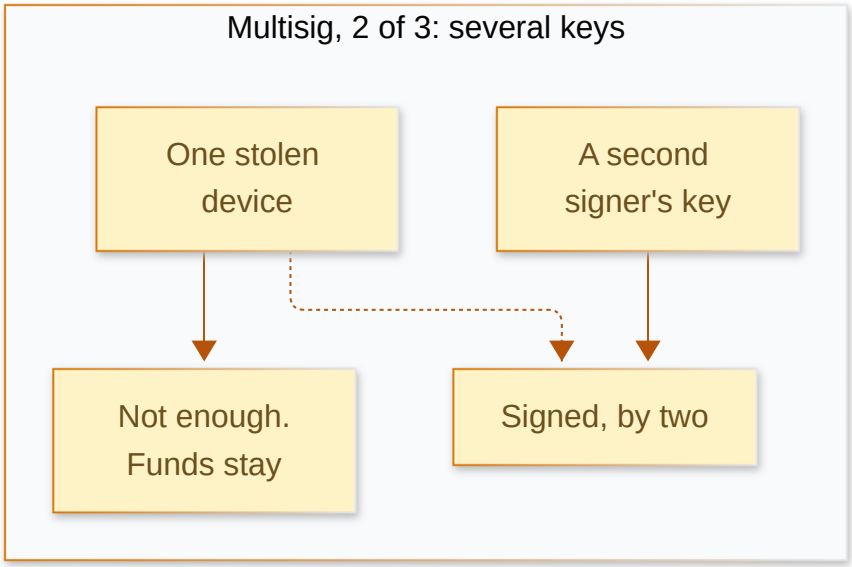
A seed phrase is the wallet. Whoever reads it owns everything behind it, from anywhere, without touching your device. So it is the one thing you never photograph, never store in the cloud, and **never type into a website**. Trezor's own warning is to enter it only as your device instructs, and "never type it into a website, form, app or message". A site asking for your words is a thief, however official it looks. This holds whatever the page, the email, the support agent or the helper says.

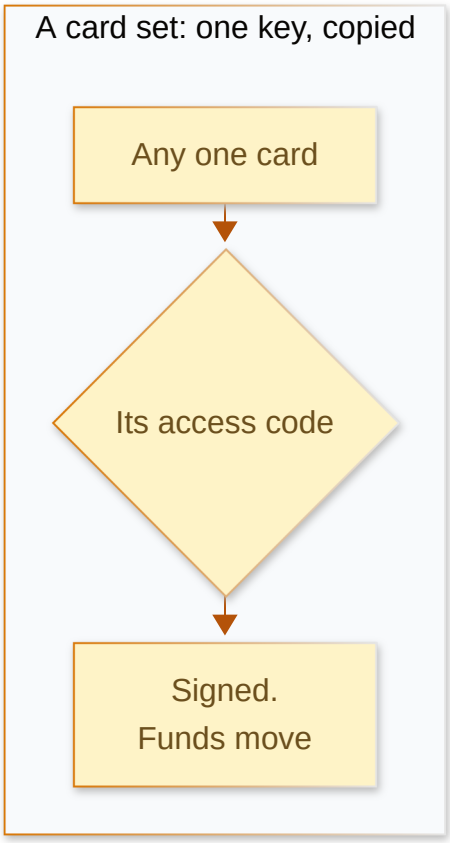
Seedless card wallets, such as Tangem

The plain fact. Tangem's backup cards are copies, and each can sign alone. A stolen card is guarded only by its access code. This is not multisig.

What Tangem's own pages say	Page, as read Nov 5, 2026
Each card receives a copy of the same private key, and all two or three give identical access.	Security features (Feb 23, 2026)
Any one card gives full access to the wallet.	Multi-account vs multiple wallets (Sep 22, 2026)
A found or stolen card can't reach the funds without the access code.	Lost card (Jun 2, 2025)
After six wrong tries, delays grow, up to 45 seconds, and reset on a correct entry. The code has a minimum of 4 characters and no maximum.	Security features
Resetting a forgotten code takes two cards from the set, so one stolen card can't reset it. The newer wallet lets you turn that reset off, and then a forgotten code can't be reset at all.	Reset a code (Apr 10, 2024)
You can't add a card to an existing wallet. If you suspect a lost card is compromised, Tangem suggests moving to a new wallet. It also says never to keep all your devices in one place.	Lost card
Set up without a seed, losing every card means losing access for good. A seed is an option on second-generation wallets and the Ring.	Lost card, Seed FAQ

Tangem's help-centre pages showed only navigation to our reader on Nov 5, so these come from its blog and learning hub. **None of the pages we read uses the word multisig about the cards.** The conclusion below is ours, drawn from their words.





What follows from that, in two depths.

	In plain words	Technically
Copies, not parts	Three cards are three whole keys, not three pieces. More cards protect you from loss, and give a thief more cards to find.	Redundancy is not a threshold. On-chain, one key signs, and nothing about the transaction shows a second approval.
The code is the whole guard	Pick a long one. A short code and a stolen card are all that stands between a thief and the wallet.	The delay slows guessing, and Tangem's pages describe no wipe or lockout. The minimum of 4 characters is not a recommendation. Whether a card resists other attacks is beyond what these pages show, so **UNVERIFIED** here.
Two cards together	Keep the cards apart. Two taken at once may be enough to get round the code.	Since a reset takes two cards from the set, two stolen together could reset the code, unless that reset was turned off. Tangem's page doesn't spell this out; it follows from the procedure.
A lost card is a decision	If one may be in someone's hands, the wallet has a spare key out there forever.	You can't revoke it or add a new card. The way out is a new wallet and moving the funds, Tangem's own suggestion.

Each of these designs is a real choice. Cards mean no words to find on a shelf and a simple recovery. Multisig means more to set up and more ways to get it wrong. The point is only to know which one you have.

Four failures, not one

"Keep it safe" means nothing until you say safe from what. A plan that handles one of these may do nothing for the next.

Failure	What it takes	What survives	What people get wrong
Fire	everything in one building, and the phone, the Mac and the drive at once	a copy elsewhere; maybe a rated safe, if the rating matches what's in it	one safe at home, and a paper rating for a drive
Flood	floor-level things, paper, a safe that isn't water-rated	a copy upstairs or elsewhere; paper sealed against water	assuming fireproof means waterproof. Check each rating separately
Theft	whatever can be carried, including a small safe. **It also reads:** a seed on paper is the wallet, and a card comes with a guess at its code	what's elsewhere and useless without something the thief lacks	a written code stored with the card; two cards in one drawer; a label saying what it is
"I lost it"	one item, quietly, and you may not know when	whatever you check on a schedule	finding out only on the day you need it. Also forgetting a code, a passphrase or where you put things, and being the only person who knows

The honest tension. More copies protect you from loss and expose you to theft. Fewer copies do the reverse. A multisig or a seed with a passphrase is one way of easing that, since a single stolen item isn't enough. It costs more care in exchange.

Two places, not one safe

- **Two places, far enough apart that one bad night doesn't reach both.** One at home, and one away: a bank box, or a sealed envelope with someone you trust who isn't already one of your recovery contacts. SOP-COM-002 says the same for Apple's key.
- **A safe rated for paper isn't rated for media.** Fire ratings measure how hot the inside may get. A retailer's guide to the UL classes puts paper at under 350°F (Class 350) and digital media at under 125°F (Class 125) (Safe and Vault Store, fire ratings, read Nov 5, 2026). So paper can come through a heat well above what that guide gives for media. We read a seller's summary, as UL 72's own text wasn't open to us, so read the label on the actual product. **A drive, a card or a hardware wallet in a paper-rated safe has no promise behind it.** Whether any given card survives a fire is **UNVERIFIED**.
- **Don't store the lock with the key.** A code doesn't go on or beside its card, and a passphrase doesn't go on or beside its seed.
- **Label nothing with what it is.** Your private list says what is where. It isn't kept with the things it lists, and it names no amounts.
- **Someone else may need to find it.** Decide who, and how they would know, without giving them access today. That is a job for a will and a lawyer, not this page.

Steps: an afternoon with a list

1. **List what you cannot replace.** On paper, privately, with no amounts. For each item, write down where every copy is.
2. **Read across the four failures.** For each item ask: what's left after a fire, after a flood, after a theft, and if I lost this one thing? *You should see:* an answer for all four, or a gap you now know about.
3. **Mark the loops.** Does anything need itself to be recovered? SOP-COM-002 has the test.
4. **Close the worst gap first.** For files, proposal 019's phases. For a key, a second place.
5. **Write down what kind of wallet you hold,** in the words above: seed, hardware, multisig, or copied cards. What can a single stolen item do?
6. **Put the test on the calendar.**

Test the recovery

A copy you have never restored from is a hope.

What	The test	How often
A backup drive	restore one file you can recognise, and open it	yearly, and after any change
An encrypted image	open it from the password on paper, not from a keychain (SOP-IT-008)	yearly
Paper keys and seeds	still there, still readable, seal intact. Don't take it out to read it near a camera or a helper	yearly
A seed phrase, before trusting it	restore on your own device, offline, as its maker instructs, before you put in anything that matters. Check that the addresses match	once, at setup
Each card in a set	use each one for a small transaction of your own, with its code, so you know each card and each code work	at setup, then yearly
A multisig	sign a small transaction using the keys you'd have left after losing one	at setup, then yearly

What never happens here

- A seed is never typed into a website, a form, a chat or an AI helper, whoever asks and whatever it says is at stake. Nobody real needs it.
- No seed, code or key is ever photographed, screenshotted, emailed or stored in a repository.
- This page names no one's holdings, wallets or hiding places, and neither should anything built from it.

Sources and related

Read on **2026-11-05**, each dated in the text where its age matters: Tangem's security features, lost card, reset a code, access code, before buying, seed FAQ and multi-account; BIP-39; Ledger Academy on multisig; Trezor on multisig and on the wallet backup; Safe and Vault Store on fire ratings. Vendors write about their own products, and these pages change.

Related: proposal 019 (*Backed up three ways*), SOP-COM-002, SOP-IT-007, SOP-IT-008.

Not legal, tax or investment advice. Nothing here recommends a product, a holding or an amount.

Change log

Date	Change
2026-11-05	Drafted from the vendors' own pages.

Part of the fleet's procedures. The master copies and the legend live in my-template.